

Candle Shop Point Of Sale System

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Overview:

The system we are designing, is POS system for a small Candle shop. The shop has moderate sales and regular customers in a small town. His current volume is around 50 sales a day. The store owner is a retired professional, and is pursuing his passion to produce and sell candles. However, despite the small number of sales, it is still lots for the senior to manage as he steps away from his high stress career for a more peaceful retirement. Through all the chaos of multiple spread sheets containing all informations about sales and inventory, as well as his aging memory, where it keeps failing him with his customers interactions. He decided that its time to invest in a small budget POS system that will take care of all the headache that comes with managing the store numbers and data. As he relaxes to enjoy his retirement and his day to day friendly interactions with his customers. He decided that he needs a way to remember all details about customers, like their names and last names to make the interaction warmer. As well as a way to keep track of the number of candles he has and what type (size, scent etc...), The number of sales he made, and a way to register employees' information as he steps out sometimes, and to manage the hires he employ when it gets busy.

Data outline:

Employees:

This entity stores information about the employees in the shop. So that we know the employee responsible for any sale, for issue resolutions as well as to track performances.

Relationships:

1:M One employee can process many sales, thus his id will be referenced in many sales. A sale can only have one employee.

(Implicit Many-to-Many) One employee can sell many candles, as he has many sales (or one sale) where there are many SaleItems, each reference one candle entity.

(Implicit Many-to-Many) Same way an employee can have many customers through having sales that reference the employee and different customers, as well as same customer with different employees through the sales entity.

employee_id: int, PK, not NULL — Unique ID. stores the primary key, a unique identifier for the employee.

first_name: varchar(50), not NULL stores the first name of the employee

last_name: varchar(50), not NULL stores the last name of the employee

hire_date: date, NULL stores the date when employee was hired

Customers:

This entity stores information about the customers in the shop. So that we know our customers, and track their information to know what they like and dislike, how much

they have bought and their purchase history. As well as their contact information for future targeting or promotions and names and last names to have warmer interactions

Relationships:

1:M (One-to-Many): One customer can be part of many sales over their lifetime, meaning their unique ID will be referenced in multiple sales. However, an individual sale is tied explicitly to only one customer.

(Implicit Many-to-Many): One customer can purchase many different candles. This occurs because their customer ID is linked to sales, which consist of one or more sale items tracking different physical candles.

(Implicit Many-to-Many): One customer can interact with many different employees across multiple store visits, as derived via the shared sales records.

customer_id: int, PK, not NULL Unique ID. Stores the primary key, a unique identifier for the customer.

first_name: varchar(50), not NULL Stores the first name of the customer.

last_name: varchar(50), not NULL Stores the last name of the customer.

email: varchar(100), unique, NULL Unique email constraint. Stores the customer's email address and strictly prevents duplicate accounts from being registered under the exact same email.

phone: varchar(15), NULL Stores the customer's telephone number for contact purposes.

Candles:

This entity serves as the core physical product catalog. It keeps track of the specific inventory on hand and establishes the exact product configurations available for sale by joining scent and size variations.

Relationships:

M:1 (Many-to-One): Many distinct candles can share the same scent profile (e.g., a "Vanilla" scent available in small, medium, or large shapes), but a single candle SKU must map to exactly one scent.

M:1 (Many-to-One): Many distinct candles can share the same size classification (e.g., multiple different scents poured into 8 oz jars), but a single candle SKU must map to exactly one size.

1:M (One-to-Many): One candle can be added to many different sale item lines across different customer receipts. A single line item in a receipt, however, isolates exactly one candle SKU.

candle_id: int, PK, not NULL Unique ID. Stores the primary key, a unique stock-keeping identifier (SKU) for the finished candle product.

scent_id: int, FK, not NULL Foreign Key reference. Links directly to Scents(scent_id) to specify the candle's scent category. Cannot be null.

size_id: int, FK, not NULL Foreign Key reference. Links directly to Sizes(size_id) to specify the candle's structural size category. Cannot be null.

price: decimal(6,2), not NULL Standard base price format. Stores the baseline retail cost of the product before transaction-specific discounts or taxes.

stock_qty: int, not NULL Inventory count. Stores the current volume of physical items actively sitting on store shelves or in storage.

last_updated: timestamp, not NULL tracks last time the entity was updated.

Sales:

This entity acts as the transaction header log. It registers the higher-level financial exchange event, tracking exactly when an exchange happened, who processed it, who bought it, and the total monetary amount received.

Relationships:

M:1 (Many-to-One): Many sales can be handled by a single employee, but a single sale record can explicitly reference only one responsible employee.

M:1 (Many-to-One): Many sales can belong to a single customer over time, but an individual sale receipt maps to exactly one customer.

1:M (One-to-Many): One sale must contain at least one or more line items (SaleItems) representing the products purchased. A single line item points back to exactly one parent sale invoice.

sale_id: int, PK, not NULL Unique ID. Stores the primary key, a unique receipt or invoice tracking number for the transaction event.

sale_timestamp: timestamp, not NULL Event timestamp. Stores the exact chronological date and time when the sale was completed.

total_amount: decimal(8,2), not NULL Financial total. Stores the exact final dollar amount paid at register checkout, including calculated sales tax and active promotions.

employee_id: int, FK, not NULL Foreign Key reference. Links directly to Employees(employee_id) to log the staff member handling checkout. Cannot be null.

customer_id: int, FK, not NULL Foreign Key reference. Links directly to Customers(customer_id) to log the purchasing customer. Cannot be null.

Scents:

This entity manages the shop's catalog of scent profiles. It decouples descriptive sensory information from physical inventory management, ensuring uniform naming conventions across your store website and backend tools.

Relationships:

1:M (One-to-Many): One scent profile can be utilized across many different candles of varying physical sizes. A candle product config, however, is bound to exactly one scent category.

scent_id: int, PK, not NULL Unique ID. Stores the primary key, a unique classification number for the olfactory profile.

scent_name: varchar(100), not NULL Scent name designation. Stores the commercial name of the scent profile (e.g., 'Lavender Fields', 'Teakwood & Amber').

description: varchar(255), NULL Descriptive notes. Stores customer-facing or internal metadata detailing the fragrance notes and scent characteristics.

Sizes:

This entity acts as a category master table governing the product sizing specs. It centralizes uniform dimensions, volumetric data, and label names to prevent data entry spelling anomalies.

Relationships:

1:M (One-to-Many): One size category can apply to many different individual candles across various scent lines. A specific candle is built to fit exactly one size profile.

size_id: int, PK, not NULL Unique ID. Stores the primary key, a unique numerical code for the size profile.

size_label: varchar(20), not NULL Market label. Stores the retail name of the size (e.g., 'Small Tin', 'Large 3-Wick').

volume_oz: decimal(4,1), not NULL Physical capacity. Stores the precise liquid or weight mass calculation in fluid ounces to power catalog filters.

SaleItems:

This is the transaction line item bridge entity. Because a single sale invoice can contain multiple items, and a single type of candle is sold across many separate customer visits, this table maps the Many-to-Many (M:N) relationship between Sales and Candles, capturing itemized transaction specifics.

Relationships:

M:1 (Many-to-One): Many individual line items can roll up into one parent sale receipt. However, an individual line item belongs to exactly one transaction.

M:1 (Many-to-One): Many line items across different store receipts can reference the exact same candle model, but a single row in this table isolates exactly one unique candle ID.

sale_item_id: int, PK, not NULL Unique Line ID. Stores the primary key, a unique identifier separating this specific item line from others.

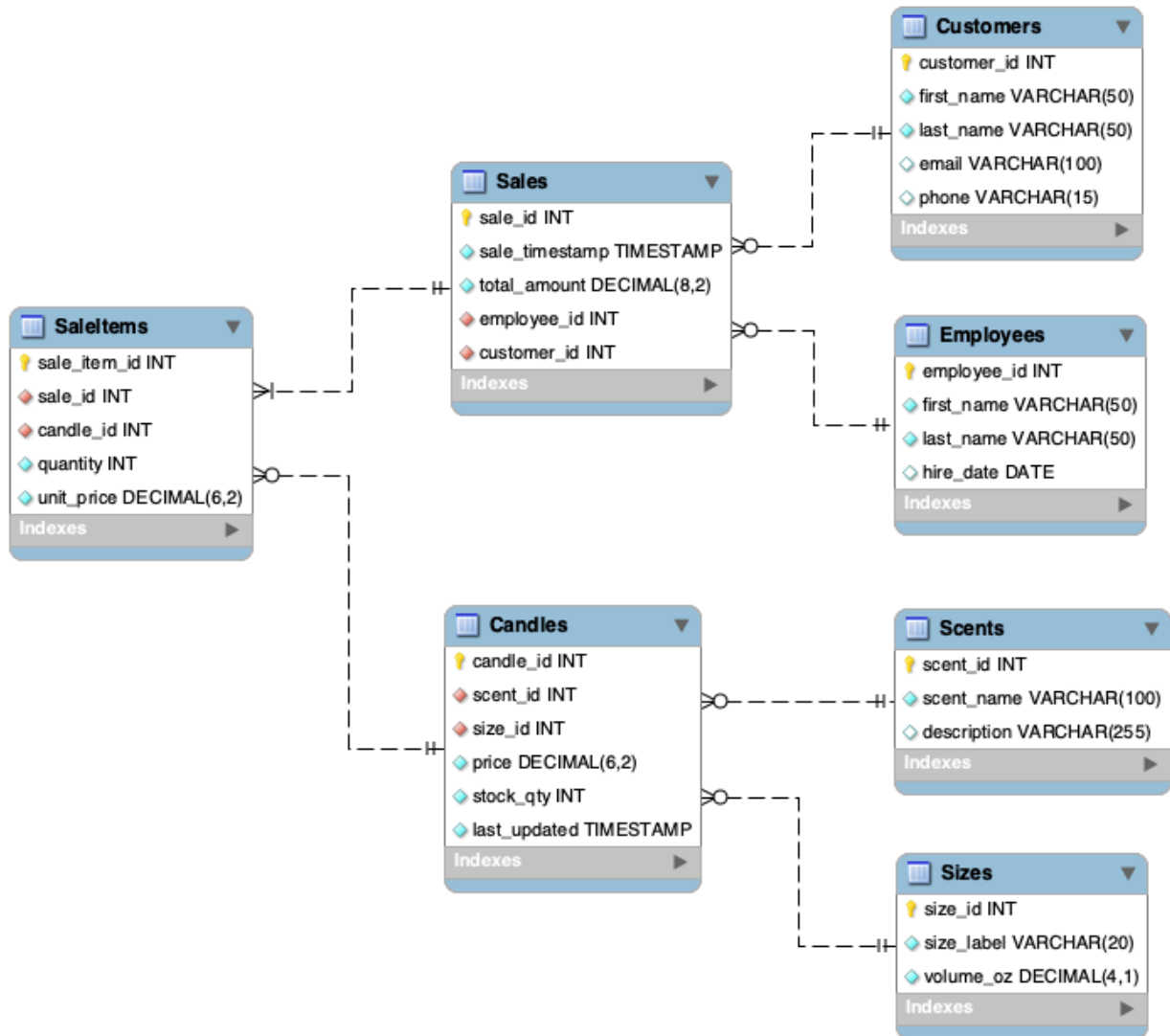
sale_id: int, FK, not NULL Foreign Key reference. Links directly to Sales(sale_id) to associate the line item with its overarching invoice receipt. Cannot be null.

candle_id: int, FK, not NULL Foreign Key reference. Links directly to Candles(candle_id) to specify exactly which candle product was bought on this line. Cannot be null.

quantity: int, not NULL Order volume. Stores the total count of this specific candle model purchased inside this specific transaction.

unit_price: decimal(6,2), not NULL Locked historical price. Stores the exact cost per unit at the literal moment of purchase, preserving historical accounting records if base prices change in the Candles table later.

Entity Relationship Diagram:



Citations:

None.